EduPlatform LMS - Architecture & Tech Stack

Complete Technical Specification & Zero-Cost Infrastructure Directory

Self-Hosted | 100% Free ($0 Cost) | Production Ready

1. WEB & APPLICATION LAYER (FRONT-END & CORE WEB SERVER)

Technology	Role / Category	Why We Use It In This Project (In Simple Words)
C# & ASP.NET MVC 5 (.NET 4.8)	Core Web Framework	Runs the main web application, user routing, security pipelines, student enrollments, exam submissions, and certificates with high enterprise performance.
Razor Views (.cshtml)	Server Templating Engine	Blends HTML with C# code to render dynamic pages on the server, ensuring rapid page loads without client-side framework complexity.
Bootstrap 5 & Bootstrap Icons	Responsive UI Framework	Provides the responsive dark theme, high-contrast layouts, forms, buttons, and iconography tailored for desktop, tablet, and mobile screens.
Vanilla JavaScript & Fetch API	Client-Side Scripting	Powers real-time interactive experiences: one-by-one MCQ quiz runner, instant answer checks, live score tallies, and asynchronous AI chat.
Windows IIS (Web Server)	Native Production Server	Hosts the .NET 4.8 web app directly on Windows with worker-process recycling, kernel-mode caching, and zero license costs.

2. DATABASE & DATA STORAGE LAYER

Technology	Role / Category	Why We Use It In This Project (In Simple Words)
Microsoft SQL Server Express 2022	Relational Database (RDBMS)	Safely persists relational entities: users, credentials, courses, modules, lessons, student progress, quiz attempts, and payment records (Free forever, up to 10 GB).
Entity Framework 6 (EF6)	Object-Relational Mapping (ORM)	Bridges C# objects with relational tables, automating CRUD operations, schema migrations, and parameterized queries that protect against SQL injection.
Redis 7 (Community Edition)	In-Memory Cache & Key-Value	Stores temporary user sessions, rate-limiting tokens, and caches frequent AI responses so repeated queries return in milliseconds with zero LLM API cost.

3. AI, RAG & VECTOR INTELLIGENCE (AISERVICE MICROSERVICE)

Technology	Role / Category	Why We Use It In This Project (In Simple Words)
Python 3.10+ & FastAPI	Async AI Microservice	Dedicated high-speed Python service dedicated to vector search, document indexing, and generative AI orchestration without blocking the main web app.
Google Gemini API (Flash Lite / 2.5)	Primary LLM Engine	Powers the Socratic Tutor, Exam Prep & Rubrics, Course Assistant, Summarizer, and 5-Question Practice Quiz Generator on Google's generous free tier.
Groq API (Llama 3.3 70B)	Failover / Backup LLM	Provides sub-second, zero-cost fallback responses if Google Gemini experiences rate-limits or network interruptions, ensuring 100% uptime.
Qdrant Vector Database	Open-Source Vector DB	Stores 3072-dimensional vector embeddings of course materials and syllabi, enabling deep semantic concept search beyond literal keyword matching.
Rank-BM25 & RRF Fusion	Hybrid Search Algorithm	Merges exact lexical keyword matching (BM25) with vector semantic search (Qdrant) via Reciprocal Rank Fusion for maximum academic retrieval accuracy.
PyMuPDF (fitz) & pdfplumber	PDF Text & Layout Extractors	Rapidly parses complex multi-column textbook PDFs, slide decks, and syllabi into clean markdown chunks ready for embedding.

4. BACKGROUND PROCESSING & QUEUE INFRASTRUCTURE

Technology	Role / Category	Why We Use It In This Project (In Simple Words)
RabbitMQ 3 (Community Edition)	AMQP Message Broker	Queues heavy asynchronous background tasks (such as large course textbook parsing and bulk vector indexing) to prevent web worker thread starvation.
Dead-Letter Queue (DLQ / DLX)	Fault-Tolerance Pipeline	Catches malformed, password-protected, or timed-out document files during ingestion, storing them for inspection without crashing worker nodes.

5. PAYMENTS, SECURITY, CERTIFICATES & GATEWAY

Technology	Role / Category	Why We Use It In This Project (In Simple Words)
Stripe API & Webhooks	Payment Gateway	Manages secure PCI-compliant card transactions in both test sandbox and live production, triggering verified course unlock webhooks.
Cloudflare Tunnels (cloudflared)	Zero-Trust Network Tunnel	Securely exposes the local Windows IIS instance to the public internet (custom domain) without opening inbound router ports or paying for static public IPs.
Cloudflare Free Tier WAF & SSL	Edge Security & CDN	Delivers free universal SSL certificates (HTTPS), DDoS mitigation, edge caching, and Web Application Firewall rules at zero cost.
BCrypt.Net & Role-Based Security	Security & Auth Enforcement	Performs adaptive salted hashing for passwords and enforces strict role-based authorization for Students, Instructors, and System Administrators.
Rotativa / wkhtmltopdf	Document Generation Engine	Renders HTML/CSS templates into pixel-perfect PDF billing receipts and verifiable, tamper-evident graduation certificates.

6. DEVOPS, CONTAINERIZATION & TELEMETRY

Technology	Role / Category	Why We Use It In This Project (In Simple Words)
Docker Desktop (WSL 2)	Container Orchestration	Runs Redis, Qdrant, RabbitMQ, and Grafana in unified Linux containers with health checks, local volume persistence, and one-command boot.
Prometheus OSS	Metrics Collection Engine	Continuously samples RAM consumption, CPU utilization, RAG pipeline latency, and endpoint throughput.
Grafana OSS	Monitoring Dashboard	Visualizes real-time performance graphs, student traffic curves, database pool utilization, and confirms $0.00 cloud spending.
k6 by Grafana	Load & Stress Testing Suite	Executes graduated stress scenarios (10 → 25 → 50 → 100 concurrent virtual users) to prove sub-200ms latency SLAs under heavy exam traffic.

Project: EduPlatform LMS (ASP.NET MVC 5 + Hybrid RAG AI Service)
Deployment Model: 100% Free / Self-Hosted Production Stack
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